

Personal Financial Assistant App Overview

<https://gemini.google.com/app/3d31fbbe80845368>

User prompt: we are talking about 3 test cases but how will we manage cards for everyone, how will we generate card of anyone

Response: You generate cards for anyone using a **parameterized rules template engine** rather than hardcoding static profiles.

Every borrower's card pulls dynamically from three runtime variables produced by your underwriting pipeline:

1. **Dynamic Numbers:** The output values (O_2, O_3, O_4) calculated for their specific income, expenses, and term.
2. **Dynamic Strategy Tags:** Conditional flags evaluated from their answers (e.g., has_prime_cibil, has_collateral, has_distress_triggers, has_productive_utility).
3. **Product Benchmark Matrix:** A deterministic lookup table defining rate caps, fee ceilings, and product categories across Indian lending markets.

The Universal Negotiation Blueprint

Regardless of who applies, the engine maps their answers into four standard sections on the card:

```
[User Answers]
├──> Evaluate Strategy Flags (Collateral? High CIBIL? Digital Debt?)
├──> Run Math Engine (Calculates Fair Rate, True APR, Safe EMI Ceiling)
└──> Assemble Negotiation Card via Modular Template:
    ├── Target Product & Caps (Derived from best-fit loan category)
    ├── Leverage Bullets (Assembled from positive borrower flags)
    ├── Traps to Avoid (Assembled from risk & category flags)
    └── Spoken Script (Dynamic string interpolation)
```

The Dynamic Assembly Engine in Python

Here is the exact production-ready pattern for engine/negotiation.py that handles any arbitrary applicant:

```
def build_negotiation_card(profile: dict, calculated: dict) -> dict:
    """
    profile: Raw user inputs (e.g., income, category, cibil, collateral_value, etc.)
    calculated: Outputs from formulas (fair_rate_min, fair_rate_max, apr, safe_emi_cap)
    """
    leverage = []
    traps = []

    # -----
    # 1. Product Routing & Core Targets
    # -----
    if profile.get("collateral_value", 0) > 0:
        target_product = "Loan Against Property (LAP) / Secured Line"
        rate_rationale = f"pledging unencumbered property worth ₹{profile['collateral_value']:,.}"
        max_fee_pct = "1.0% + GST"
    elif profile.get("is_productive_asset", False):
        target_product = "Commercial Asset / Priority Sector Loan"
        rate_rationale = f"financing a productive asset yielding ₹{profile.get('monthly_revenue_gain', 0):.}/mo"
        max_fee_pct = "1.5% + GST"
    else:
        target_product = "Standard Personal / Installment Loan"
        rate_rationale = "my verified income stability and FOIR debt headroom"
        max_fee_pct = "1.0% to 1.5% + GST"

    # -----
    # 2. Dynamic Leverage Generator (Rule-Based Modifiers)
    # -----
    # CIBIL leverage
    cibil = profile.get("cibil")
    if cibil and cibil >= 750:
        leverage.append(f"Credit score of {cibil} places me in the prime pricing tier.")
    elif not cibil:
        leverage.append("Demonstrated positive cash flow offsets unscored bureau status.")

    # Cash flow leverage
    if calculated["safe_emi_cap"] > profile.get("existing_emi", 0) * 1.5:
        leverage.append(f"Healthy unencumbered cash buffer with ₹{calculated['safe_emi_cap']:,.}/mo safe repayment ca")

    # Vintage leverage
    if profile.get("business_vintage_years", 0) >= 3 or profile.get("job_tenure_years", 0) >= 3:
        leverage.append("Over 3 years of verified operational/job stability at the same institution.")
```

```
# -----
# 3. Dynamic Trap Warning Generator
# -----
if profile.get("has_app_loans", False) or profile.get("bounces_last_6m", 0) > 0:
    traps.append("Reject any new non-productive loan prior to clearing active high-cost debt.")

traps.append(f"Decline processing fees exceeding {max_fee_pct}.")
traps.append("Refuse bundled credit-life or single-premium insurance deducted from disbursement.")

if target_product.startswith("Loan Against Property"):
    traps.append("Do not let the lender divert you to an unsecured business loan at >16% APR.")

# -----
# 4. Spoken Script Formulation (String Interpolation)
# -----
spoken_script = (
    f"I am applying for a {target_product}. "
    f"Based on {rate_rationale}, standard regulatory pricing indicates a fair rate of "
    f"{calculated['fair_rate_min']}% - {calculated['fair_rate_max']}% with an all-in APR cap of {calculated['apr']}%",
    f"My monthly commitment ceiling is strictly ₹{calculated['safe_emi_cap']:,}/month; "
    f"I will not accept structured tenures or insurance add-ons that breach these limits."
)

return {
    "target_product": target_product,
    "fair_rate_band": f"{calculated['fair_rate_min']}% - {calculated['fair_rate_max']}%",
    "apr_ceiling": f"{calculated['apr']}%",
    "monthly_emi_ceiling": f"₹{calculated['safe_emi_cap']:,}/mo",
    "processing_fee_cap": max_fee_pct,
    "leverage_points": leverage,
    "traps_to_reject": traps,
    "spoken_script": spoken_script
}
```

How This Works in Practice Across Different Users

- **A Government Teacher (Salaried, CIBIL 720, No Property):**
 - *Product:* Prime Personal Loan
 - *Leverage:* PSU/Govt employment stability, low risk of default.
 - *Script:* Points out stable public sector salary to negotiate rates under 11.5%.
- **A Dairy Farmer / Agri-trader (Self-Employed, High Cash, Land/Shop Owned):**
 - *Product:* Secured Agri/LAP Line
 - *Leverage:* Real estate backing, zero uncollateralized default risk.
 - *Script:* Demands secured rates of 9.5%–10.5% and refuses unsecured NBFC debt.
- **An Independent Freelance Designer (Informal/Gig, No App Debt, Good Cash Flow):**
 - *Product:* Professional Working Capital / Equipment Line
 - *Leverage:* Clean banking credits, productive design equipment ROI.
 - *Script:* Rejects consumer loan pricing and insists on commercial equipment financing rates.

The template logic remains identical across every user; only the evaluated flags and computed figures change. This keeps your application deterministic, explainable, and fully compliant with the challenge rules.

User prompt: as per my knowledge i have completed 4 portions the categories,their question and how they will contribute ,how to calculate the verdict and the logic behind negotiation card, whats left to figure out?

Response: To finalize the project and meet every rubric requirement from the brief, there are **5 critical portions left to figure out and build:**

1. The "Every Number Has a Why" Sentence Generator

- **The Requirement:** Rule 4 states: *"The borrower must be able to read, in one sentence, why the ceiling is ₹22,000 and not ₹30,000."*
- **What you need to engineer:** A templated string builder in your engine that pairs every output (O_1, O_2, O_3, O_4) with an exact plain-English diagnostic sentence explaining which constraint bounded the number (e.g., FOIR threshold, rent deduction, or high-risk spread markup).

2. The 3 Mandatory Persona Run-Throughs

- **The Requirement:** Deliverable #3 mandates complete end-to-end run-throughs for **Priya, Ravi, and Anita**.
- **What you need to build:** Hardcoded test profiles or test vectors validating:

- **Priya (Salaried, Bengaluru):** Prime personal loan pricing, low APR, Borrow verdict.
- **Ravi (Self-Employed, Mysuru):** Product pivot to LAP using unencumbered property value, avoiding predatory unsecured rates.
- **Anita (Informal, Hubballi):** Triggering the "Don't Borrow" kill-switch due to recent bounces and >30% app loans, followed by a restructuring pathway for EV financing.

3. The Stress-Testing Mathematical Scenarios (O_4)

- **The Requirement:** Evaluating how safe borrowing thresholds behave under shocks.
- **What you need to code:** Explicit stress-test logic modeling:
 - An income shock: 20% drop in take-home cash.
 - An interest rate shock: +2.0% rate spike on floating loans.
 - The threshold condition: whether the total debt obligation exceeds **65% of stressed net income**.

4. Documentation Deliverables (RULES.md & Methodology)

- **The Requirement:** The submission requires a standalone document detailing the financial reasoning and assumptions.
- **What you need to write:**
 - Benchmark assumptions for Indian lending (base FOIR percentages, statutory GST rates on processing fees, product rate bands).
 - Explanation of how "Unknown" inputs are modeled without defaulting to zero (satisfying Rule 3: "Unknown is never zero").
 - Step-by-step documentation of the confidence scaling logic.

5. The User Interface & Architecture Implementation

- **The Requirement:** A zero-API, zero-database local application (e.g., Streamlit, Pure HTML/JS, or Flask) with clean UI-to-engine separation.
- **What you need to wire together:**
 - **Adaptive Question Wizard:** Dynamically rendering the next question based on the selected category without page reloads.
 - **Interactive Confidence Bar:** Visual bar scaling from 60% to 95% using the mapped Excel lookup table.
 - **Results Dashboard:** Clean card layout displaying O_1 to O_4 , the one-sentence explanations, and the mobile-ready Negotiation Card.

Which of these would you like to tackle next: writing the **"Every Number Has a Why" string generator**, or setting up the exact test runs for **Priya, Ravi, and Anita**?

User prompt: explain point 1

Response: Rule 4 from the guidelines explicitly dictates:

"Every number has a why. The borrower must be able to read, in one sentence, why the ceiling is ₹22,000 and not ₹30,000."

Lending calculators typically display a final number (like a sanction amount or an EMI limit) without context. To fulfill this requirement, every output (O_1 to O_4) generated by the engine must be accompanied by a single, dynamically formatted English sentence explaining the exact mathematical bottleneck or policy rule that produced it.

The Bottleneck Concept: Why Numbers Stop Where They Do

Every output is governed by competing constraints. The sentence generator must identify which specific variable acted as the **limiting factor**:

- For the **Safe EMI Ceiling (O_4)**, the limit is either bounded by the **bank's FOIR limit** or by the borrower's **actual living expenses and emergency buffer**.
- For the **Fair Rate Band (O_3)**, the spread is dictated by **credit score status**, **unsecured vs. asset-backed collateral**, or **risk penalties**.
- For the **Borrowing Amount (O_2)**, the figure reflects the gap between **what a bank allows** and **what safe cash flow permits**.
- For the **Final Verdict (O_1)**, the decision explains whether cash flow was sufficient, an active default occurred, or leverage was stretched too thin.

How the Four Sentences Are Generated

1. Output 4 Explanation (Why the Monthly EMI Ceiling is ₹X and not ₹Y) The engine checks whether the safe ceiling was limited by external bank policy or home expenses.

- *If bounded by living expenses:*

"Why: While standard bank FOIR allows up to ₹35,000/month, your ₹28,000 rent and ₹18,000 household living costs cap your safe payment at ₹22,000 to keep a 10% emergency reserve intact."

- If bounded by existing debt obligations:

"Why: Your safe repayment capacity is capped at ₹14,000 because your existing car EMI of ₹16,000 already utilizes over 30% of your net monthly take-home."

2. Output 3 Explanation (Why the Rate is X% and not Y%) The engine identifies the primary driver of the interest rate spread.

- Prime Salaried Profile:

"Why: Your 780 CIBIL score and Tier-1 corporate employer qualify you for the lowest unsecured personal loan spread (10.5%–11.2%), but standard bank processing fees of 1.5% + GST adjust your true all-in APR to 11.9%."

- Secured Product Pivot:

"Why: Pledging an unencumbered shop premises shifts your loan category to Loan Against Property (LAP), reducing the interest band from standard 18%+ business loan pricing down to 9.5%–10.5%."

- Unscored / Unknown Bureau:

"Why: Because your credit score is unverified, the engine models a wider rate band of 12.5%–15.5% without penalizing you as a defaulter, reflecting standard unrated risk."

3. Output 2 Explanation (Why the Lender Limit and Safe Limit Differ) Explains the divergence between bank underwriting and personal financial safety.

- Typical salaried profile with rent:

"Why: Bank underwriting formulas approve up to ₹14,50,000 based on gross salary, but taking more than ₹9,20,000 would leave you with insufficient cash flow to cover your essential living costs during an emergency."

4. Output 1 Explanation (Why the Verdict is Borrow, Borrow Less, or Don't Borrow) Communicates the primary risk assessment.

- Verdict: BORROW LESS

"Why: Your requested loan of ₹12,00,000 requires a monthly EMI of ₹26,500, which exceeds your maximum safe monthly payment ceiling of ₹20,000 by ₹6,500."

- Verdict: DON'T BORROW

"Why: You have had a loan bounce in the last 6 months and hold active instant app debt at over 30% APR; adding new debt before clearing these obligations risks severe default."

Implementation in Python (engine/explanations.py)

In code, this is handled via deterministic conditional templates:

```
def explain_emi_ceiling(profile: dict, calculated: dict) -> str:
    lender_max_emi = calculated["lender_max_emi"]
    safe_emi = calculated["safe_emi_cap"]
    rent = profile.get("rent", 0)
    living = profile.get("living_expenses", 0)

    if safe_emi < lender_max_emi:
        return (
            f"Why: While bank FOIR norms permit up to ₹{lender_max_emi:,}/month, "
            f"your monthly housing outflow of ₹{rent:,} and living expenses of ₹{living:,} "
            f"cap your safe payment at ₹{safe_emi:,} to protect your living buffer."
        )
    else:
        return (
            f"Why: Your safe repayment capacity is capped at ₹{safe_emi:,}/month "
            f"because regulatory FOIR thresholds prevent total debt obligations from exceeding 50% of your income."
        )

def explain_rate_band(profile: dict, calculated: dict) -> str:
    if profile.get("collateral_value", 0) > 0:
        return (
            f"Why: Pledging unencumbered property worth ₹{profile['collateral_value']:,} "
            f"secures prime LAP pricing of {calculated['fair_rate_min']}%–{calculated['fair_rate_max']}%, "
            f"avoiding high-cost unsecured credit."
        )
    elif profile.get("cibil", 0) >= 750:
        return (
            f"Why: A strong CIBIL score of {profile['cibil']} places you in the prime borrower bracket, "
            f"unlocking the lowest market spread with an all-in APR of {calculated['all_in_apr']}%."
        )
    else:
        return (
            f"Why: A wider rate band of {calculated['fair_rate_min']}%–{calculated['fair_rate_max']}% "
            f"is applied because your credit history is unscored, ensuring transparency without defaulting to zero."
        )
```

This ensures your application adheres to Rule 4 by translating calculations into clear, plain-language statements directly below each result.

User prompt: so basically there will be two section for ouputs ,,one will be giving every numbers and another will be why the number right?

Response: Yes, conceptually you have the **hard figures** and the **underlying reasons**, but you should **not** separate them into two completely disconnected sections or pages on your UI.

The guidelines state under Rule 4:

"Every number has a why. The borrower must be able to read, in one sentence, why the ceiling is ₹22,000 and not ₹30,000."

If you put all numbers in one section and hide the explanations in a separate section below, the borrower has to cross-reference back and forth, breaking the clarity the judges are evaluating.

Instead, present them as **paired result cards** where each calculated fact is immediately followed by its single-sentence explanation.

Recommended Output Card Layout

Card 1: Final Recommendation (O_1)

- **The Result: VERDICT: BORROW LESS**
- **The Why:** "Why: Your requested loan of ₹12,00,000 requires a monthly EMI of ₹26,500, which exceeds your safe monthly payment ceiling of ₹20,000 by ₹6,500."

Card 2: Borrowing Limit (O_2)

- **The Result: Lender Limit: ₹14,50,000 | Safe Borrowing Limit: ₹9,20,000**
- **The Why:** "Why: While standard bank FOIR allows up to ₹14.5L, your ₹28,000 rent and household living costs cap your safe repayment capacity at ₹9.2L to maintain essential liquidity."

Card 3: Fair Interest & Cost (O_3)

- **The Result: Fair Rate: 10.5% – 11.25% (All-In APR: 11.9% with fees)**
- **The Why:** "Why: Your 780 CIBIL score qualifies you for prime unsecured corporate pricing, but standard bank processing fees of 1.5% + GST adjust your true annual borrowing cost to 11.9%."

Card 4: Monthly Payment Ceiling & Stress Resilience (O_4)

- **The Result: Monthly EMI Ceiling: ₹22,000/month (Stress-Tested)**
- **The Why:** "Why: Even if your net monthly income drops by 20% to ₹88,000, paying ₹22,000 keeps total debt obligations under our 50% safety threshold."

Directly below these four paired output cards, place the **Negotiation Card** as the tactical takeaway for the bank branch. This keeps the visual hierarchy clean, scannable, and directly aligned with the brief's requirement that every number carries its reason alongside it.

User prompt: no no they will be in the same plasce i was just asking that the output will be 2 folded right

Response: Yes, exactly. Every output is two-folded:

- **Fold 1 (The Quantitative Fact):** The hard calculation produced by the mathematical formula (e.g., ₹22,000/mo, 10.5% – 11.25%, or Borrow Less).
- **Fold 2 (The Qualitative Grounding):** The exact one-sentence diagnostic explaining the bottleneck or driver behind that specific figure (e.g., "Why: Your rent of ₹28,000 caps your cash flow...").

In data structure terms, your output engine should return an object where every key contains both pieces paired together:

```
outputs = {
  "verdict": {
    "value": "BORROW LESS",
    "why": "Your requested loan requires an EMI of ₹26,500, exceeding your safe surplus by ₹4,500."
  },
  "borrowing_limits": {
    "lender_max": 1450000,
    "safe_limit": 920000,
    "why": "Banks approve on gross salary, but living expenses limit your safe principal to ₹9.2L."
  },
  "fair_rate": {
    "band": "10.5% - 11.25%",
    "all_in_apr": "11.9%",
    "why": "Your 780 CIBIL unlocks Tier-1 pricing, while a 1.5% fee plus GST sets the APR at 11.9%."
  },
  "monthly_ceiling": {
    "value": 22000,
    "why": "A 20% drop in take-home pay still leaves enough cash flow to service this payment safely."
  }
}
```

This keeps the engine mathematically rigorous while directly satisfying the rule that every single number carries its "why".

User prompt: the second fold will be calculated based on which factors?

Response: The second fold (the "Why" sentence) is calculated by checking **which specific variable acted as the active constraint or dominant driver** in that formula.

In code, this is handled through deterministic **bottleneck comparison logic**:

1. Factors Deciding the Why for Output 4 (Monthly EMI Ceiling)

The engine compares two competing constraints:

Lender Max EMI (FOIR Limit) vs. Borrower Safe EMI (Residual Cash Flow)

- **Deciding Factor 1: High Fixed Living Outflows (Rent / Expenses)**
 - *Trigger*: If Safe EMI < Lender Max EMI.
 - *Active Factor*: The borrower's declared rent or non-negotiable living expenses are the bottleneck, not the bank's rules.
 - *Generated Why*: Flags the exact rupee deductions (rent + living costs) that pulled the ceiling below the bank's allowable limit.
- **Deciding Factor 2: High Pre-Existing Debt Burden (E_{old})**
 - *Trigger*: If existing loan payments already eat up $\geq 30\%$ of net monthly income.
 - *Active Factor*: Existing debt servicing leaves minimal debt capacity under FOIR guidelines.
 - *Generated Why*: Identifies ongoing EMIs as the primary constraint holding back the ceiling.
- **Deciding Factor 3: Stress-Test Threshold Breach**
 - *Trigger*: If the payment survives normal cash flow but exceeds **65% of net income under a 20% drop simulation**.
 - *Active Factor*: Lack of cash buffer under income shock.
 - *Generated Why*: Explicitly notes that the ceiling was reduced to preserve safety during an income downturn.

2. Factors Deciding the Why for Output 3 (Fair Rate & APR Band)

The rate why-sentence is driven by **risk-tier qualification, collateral presence, and fee drag**:

- **Deciding Factor 1: Asset Security Pivot (Collateral)**
 - *Trigger*: Unencumbered real estate / shop value entered > 0 .
 - *Generated Why*: Notes that pledging clear property shifts the loan to secured LAP pricing (9.5%–10.5%), bypassing unsecured business loan rates.
- **Deciding Factor 2: Bureau Score Tier (CIBIL)**
 - *Trigger*: CIBIL ≥ 750 vs. CIBIL < 700 vs. "Unknown".
 - *Generated Why*: States whether prime bureau standing earned a rate discount or unverified bureau history forced a wider risk band.
- **Deciding Factor 3: Fee Drag (IRR Impact)**
 - *Trigger*: Difference between nominal interest rate and true all-in APR ($\Delta = \text{APR} - \text{Nominal Rate}$).
 - *Generated Why*: Quantifies the exact fee drag (e.g., bank processing fees of 1.5% + 18% GST adding 0.6% to the real annual cost).

3. Factors Deciding the Why for Output 2 (Lender Sanction vs. Safe Limit)

This sentence is decided by the **spread between gross regulatory allowance and personal disposable cash flow**:

- **Deciding Factor: Income Allocation Disparity**
 - *Calculation*: Checks the delta: Lender Sanction — Safe Borrowing Limit.
 - *Deciding Variables*: The bank evaluates gross in-hand income using standard 50% FOIR without accounting for personal rent, dependants, or savings buffers.
 - *Generated Why*: Highlights the gap between what a bank will recklessly lend versus what the borrower can borrow without depleting their living buffer.

4. Factors Deciding the Why for Output 1 (The Verdict)

The verdict sentence is decided by a strict **hierarchical triage sequence**:

Priority 1: Distress / Kill-Switch Factors

- └ Active loan/auto-debit bounces in last 6 months? → "Why: Active default history detected"
- └ Carrying fintech debt at >30% APR? → "Why: High-cost predatory debt trap"

Priority 2: Over-Leverage Factors

- └ Requested EMI > Safe EMI Ceiling? → "Why: Payment exceeds safe monthly surplus by ₹X"

Priority 3: Affordability Factors

- └ Requested EMI ≤ Safe EMI Ceiling? → "Why: Payment fits within disposable cash buffers"

- **Kill-Switch Flagged:** Explains why immediate debt consolidation or restructuring must precede new loans.
- **Excess Leverage Flagged:** Pinpoints the exact rupee amount by which the requested loan's EMI overshoots disposable cash.
- **Clean Approval:** Confirms that monthly repayment leaves essential living expenses and emergency buffers intact.

User prompt: how we will decide which reason to show?

Response: To select the single correct reason from multiple potential variables, the engine uses **Deterministic Priority Trees** (also known as binding constraint cascades).

Instead of guessing or combining five points into a messy paragraph, the engine evaluates the user's financial profile against an ordered list of conditions. The **first rule that gets triggered becomes the active reason** shown on the card.

1. Decision Logic for Output 1: The Verdict Reason

The verdict reason follows a strict risk hierarchy. System distress always takes precedence over simple budget math.

Check 1: Are there active bounces or defaults in the last 6 months?

- └ YES → Reason: "Why: An active payment bounce flags immediate default distress; new debt will deepen insolven

Check 2: Is the user carrying predatory app debt (>24% APR)?

- └ YES → Reason: "Why: You carry high-cost digital app debt (>24% APR); taking new loans before consolidating t

Check 3: Is Safe EMI ≤ 0 (Existing expenses already exceed income)?

- └ YES → Reason: "Why: Essential living costs and existing debt already consume 100% of your take-home pay, lea

Check 4: Does Requested EMI exceed Safe EMI Ceiling?

- └ YES → Reason: "Why: Your requested loan requires an EMI of ₹{req_emi:}, which overshoots your safe monthly

Default (All checks pass):

- └ Reason: "Why: Your requested EMI of ₹{req_emi:} fits comfortably within your ₹{safe_emi:} safe monthly limit

2. Decision Logic for Output 4: Monthly Ceiling Reason

The monthly ceiling represents the maximum payment the borrower can afford. The reason identifies the **tightest financial bottleneck**:

```
# The engine compares the three possible ceiling drivers:
if safe_emi_cap <= 0:
    reason = f"Why: Your ongoing EMIs (₹{existing_emi:}) and living expenses (₹{living:}) leave no surplus cash."

elif stress_test_failed:
    # Triggered if EMI exceeds 65% of income after a simulated 20% drop
    reason = f"Why: Under a 20% income shock, an EMI above ₹{safe_emi_cap:} would consume over 65% of your remainin

elif safe_emi_cap < lender_foir_emi:
    # Living expenses are tighter than standard bank formulas
    reason = f"Why: While bank FOIR rules allow up to ₹{lender_foir_emi:}, your fixed rent of ₹{rent:} and living

else:
    # Bank regulatory FOIR (50%) is the tighter constraint
    reason = f"Why: Capped at ₹{safe_emi_cap:} because total debt repayments cannot legally exceed 50% of your net
```

3. Decision Logic for Output 3: Fair Rate Reason

The rate reason explains what caused the interest spread to land where it did, checking for collateral first, bureau status second, and fee impact third:

Priority 1: Collateral Backing (Largest spread mover)

- └ Condition: unencumbered_property_value > 0
- └ Reason: "Why: Pledging an unencumbered shop/property shifts you to Loan Against Property (LAP), dropping yo

Priority 2: Prime Credit Score

- └ Condition: cibil ≥ 750
- └ Reason: "Why: Your CIBIL score of {cibil} unlocks Tier-1 prime corporate spreads, while a {pf_pct}% process

Priority 3: Subprime / Credit-Hungry Profile

- └ Condition: cibil < 700 or recent_inquiries > 3
- └ Reason: "Why: High credit utilization and recent bureau inquiries add a risk markup, widening your borrowin

Default: Bureau Unscored / Unknown

└ Condition: cibil is None / "Unknown"

└ Reason: "Why: A wider band of {rate_min}%–{rate_max}% is modeled because your credit history is unverified,

4. Decision Logic for Output 2: Maximum Amount Gap Reason

This explains the dollar gap between **Lender Sanction** and **Borrower Safe Limit**:

- If the gap is large (Lender offers significantly more than is safe):

"Why: Banks approve up to ₹{lender_limit:.} based purely on gross income, but your household rent and living expenses restrict your safe debt capacity to ₹{safe_limit:.}."

- If the limits are nearly identical:

"Why: Your minimal fixed living expenses and zero existing loans allow your safe borrowing ceiling to closely match standard banking sanction limits."

Summary of Implementation

In your Python code, you don't need machine learning or complex text models. You write simple, readable if/elif/else blocks in engine/explanations.py that check these exact mathematical flags and return the corresponding string template.

User prompt: these will work even if only the mandatory qs filled??and will the answer differ if variable filled also?

Response: The decision logic is built to function with **mandatory questions only**, and the explanations will dynamically evolve and sharpen as the user answers the variable questions.

This behavior fulfills two core principles from the evaluation rubric:

1. **Rule 2:** "Confidence widens with silence... Never narrow a range you have no basis to narrow."
2. **Rule 4:** "Every number has a why."

How the Explanation Adapts: Mandatory-Only vs. Variable-Added

When only mandatory inputs are provided, the engine relies on broad assumptions and wide bands. As variable answers are supplied, the bottleneck shifts from **generic assumptions** to **verified facts**, sharpening both the numeric values and the corresponding "Why" sentences.

Output 3: Fair Rate & APR Band

- **Mandatory Questions Only:**

- Calculated Output: **11.0% – 16.5%** (Spread: 5.5%)
- Reason Generated:

"Why: A wide interest band of 11.0%–16.5% is shown because employer classification, emergency cash reserves, and collateral status remain unverified."

- **Variable Questions Answered (e.g., Salaried adds Tier-1 Employer & 6-month buffer):**

- Calculated Output: **10.5% – 11.2%** (Spread: 0.7%)
- Reason Generated:

"Why: Confirmed Tier-1 corporate employment and a 6-month liquid savings reserve tighten your pricing into the prime corporate bracket at an all-in APR of 11.8%."

Output 4: Monthly Safe EMI Ceiling

- **Mandatory Questions Only:**

- Calculated Output: **₹18,500/month** (Conservative 15% default cash cushion applied)
- Reason Generated:

"Why: Standard bank FOIR allows up to ₹35,000, but your reported ₹20,000 rent and basic living costs cap your safe payment at ₹18,500 using an unverified baseline reserve."

- **Variable Questions Answered (e.g., Self-Employed enters low-season revenue variance):**

- Calculated Output: **₹14,000/month**
- Reason Generated:

"Why: Because your retail business experiences a 35% seasonal cash drop during monsoon troughs, your safe ceiling is anchored strictly to your lowest earning month."

Output 1: Final Verdict

- **Mandatory Questions Only:**

- *Calculated Output:* **BORROW** (Tentative / Wide Bands)
- *Reason Generated:*

"Why: Your requested EMI of ₹15,000 fits within your estimated disposable surplus, assuming no unverified debt defaults or active bounces exist."

- **Variable Questions Answered (e.g., Informal earner reports 1 bounce in the last 6 months):**

- *Calculated Output:* **DON'T BORROW**
- *Reason Generated:*

"Why: A payment bounce in the last 6 months triggered our distress safety rule; adding new debt before clearing existing arrears creates high risk of complete default."

Fallback Logic in engine/explanations.py

In your engine code, handle the mandatory-only state using a boolean flag such as `is_preliminary` (or checking if `confidence_score == 60`):

```
def explain_rate_output(profile: dict, calculated: dict, is_preliminary: bool) -> str:
    if is_preliminary:
        return (
            f"Why: A wide interest band ({calculated['fair_rate_min']}%-{calculated['fair_rate_max']}%) "
            f"is applied because your employer tier, collateral, and debt history are unverified."
        )

    # If variable fields are filled, identify the exact binding factor:
    if profile.get("has_collateral"):
        return (
            f"Why: Pledging an unencumbered asset qualifies you for Loan Against Property (LAP) pricing "
            f"at {calculated['fair_rate_min']}%-{calculated['fair_rate_max']}%."
        )
    elif profile.get("cibil", 0) >= 750 and profile.get("employer_tier") == "Tier-1":
        return (
            f"Why: Your 750+ CIBIL and Tier-1 employer tier secure the prime spread "
            f"({calculated['fair_rate_min']}%-{calculated['fair_rate_max']}%), with fees fixing APR at {calculated['fair_rate_max']}%."
        )
    else:
        return (
            f"Why: Risk adjustments for credit utilization and job vintage set your rate band "
            f"at {calculated['fair_rate_min']}%-{calculated['fair_rate_max']}%."
        )
```

The app is fully functional and explains its math at **60% confidence** with mandatory answers alone. As variable answers are supplied, the numbers narrow and the text shifts from **general baseline warnings** to **specific, verified financial reasons**.

User prompt: save the whole plannings a readme file,dont save code only the planning of 1.categories2.questions and their impact percentage3.formulas and how to give verdict4.how to calculate the 2nd layer of output "the why" for the 1st ouput5.engineering behind the arguement card selection.Make it as descriptive as possible so that tommorow when i will make the app using antigravity it will guide the agent.as well as make it in such simpler terms so that i can read it and understand the process

Response: ARCHITECTURE & SPECIFICATION PLAN: DETERMINISTIC BORROWING COPILOT

1. CATEGORIES & PERSONA ARCHETYPES

The system evaluates three distinct borrower categories designed to reflect the realities of the Indian retail lending market. Each category uses fundamentally different risk metrics and product paths.

A. Salaried (Prime Corporate / Institutional)

- **Archetype Example:** Priya, software engineer at a Tier-1 MNC in Bengaluru.
- **Core Underwriting Goal:** Optimize interest pricing, minimize fee drag, and assess real discretionary cash after urban lifestyle expenses.
- **Primary Failure Mode:** Lifestyle inflation, over-leveraging via multiple credit cards and personal loans, and assuming gross bonus pay translates to safe monthly debt capacity.
- **Primary Product Track:** Unsecured Prime Personal Loans (Bank/Corporate Salary Tier).

B. Self-Employed / MSME (Merchants & Small Enterprises)

- **Archetype Example:** Ravi, runs a family kirana store in Mysuru with strong cash flow but conservative income tax returns (ITR).

- **Core Underwriting Goal:** Identify collateral to pivot the borrower away from predatory unsecured business debt into prime asset-backed structures.
- **Primary Failure Mode:** Thin formal tax returns causing mainstream banks to reject unsecured requests, pushing the borrower into merchant cash advances or high-cost NBFC business loans (>18%–24%).
- **Primary Product Track:** Secured Loan Against Property (LAP) or Micro-Mortgage.

C. Informal / Gig Economy (Platform & Daily Earners)

- **Archetype Example:** Anita, food delivery rider in Hubballi driving a rented petrol scooter, balancing volatile daily platform earnings.
- **Core Underwriting Goal:** Distress screening (detecting defaults and instant app traps) and modeling the return on investment (ROI) of productive equipment.
- **Primary Failure Mode:** Reliance on payday digital loan apps charging >30% APR, auto-debit bounces, and borrowing for pure consumption rather than earning assets.
- **Primary Product Track:** Priority Sector Commercial EV / Productive Asset Hypothecation.

2. QUESTION FRAMEWORK & CONFIDENCE PROGRESSION

The application operates on a strict rule: **Silence widens bands; information tightens bands.**

- **The Baseline (60% Confidence):** Answering the 10 Universal Mandatory questions guarantees a baseline calculation. The outputs display wide bands (e.g., rate ranges of $\pm 2.5\%$) because critical risk factors are unverified.
- **The Progression (60% to 95% Confidence):** As the borrower answers up to 18 category-specific variable questions, each answer provides a specific percentage gain on the confidence bar while narrowing output spreads.
- **Confidence Cap (95%):** Confidence never reaches 100% because self-reported inputs lack third-party bureau APIs or document verification.

Universal Mandatory Questions (10 Questions – Base 60%)

Every user must answer these 10 baseline questions to initialize cash flow calculations:

1. **Loan Purpose:** Wedding, Business Inventory, Vehicle, Renovation, Debt Consolidation, Emergency. (*Sets product bounds and default tenure ceiling*).
2. **Target Loan Amount (₹):** Principal requested (P_{req}). (*Tests requested EMI against safe limits*).
3. **Preferred Tenure (Months):** Repayment timeline (12 to 120 months). (*Sets amortization schedule n*).
4. **Primary Employment Type:** Salaried, Self-Employed, or Informal/Gig. (*Sets regulatory FOIR cap: 40%–50% and branches logic*).
5. **Net Monthly In-Hand Income (₹):** Actual verified monthly take-home earnings (I_{net}). (*Primary denominator for all affordability checks*).
6. **Existing Monthly Loan EMIs (₹):** Ongoing monthly debt obligations (E_{old}). (*Deducted from available FOIR limit*).
7. **Monthly Rent / Housing Outflow (₹):** Fixed non-negotiable shelter costs (C'_{rent}). (*Deducted directly in safe cash calculations*).
8. **Essential Living Expenses (₹):** Groceries, utilities, fees, and healthcare (C_{living}). (*Non-negotiable living floor*).
9. **Credit Score (CIBIL):** 750+, 700–749, Below 700, or Unknown/Unscored. (*Discounts rate or widens bands without defaulting to zero*).
10. **Borrower Age:** Years. (*Caps maximum tenure based on a 60-year working boundary*).

Variable Questions & Confidence Allocation (18 Per Category)

Salaried Variable Set

- **S01. Employer Corporate Tier** (+3.0% → Cum. 63.0%): Tier-1/Govt raises FOIR from 50% to 55% and drops rate spread by 0.50%; Startups cap FOIR at 45%.
- **S02. Current Employment Vintage** (+2.5% → Cum. 65.5%): ≥ 3 years narrows spread by $\pm 0.50\%$; < 1 year adds 0.75% job-hopping penalty.
- **S03. Total Career Vintage** (+2.0% → Cum. 67.5%): > 5 years unlocks 60–72 month tenures; < 2 years caps tenure at 36 months.
- **S04. Salary Credit Method** (+2.0% → Cum. 69.5%): Bank transfer validates 100% income; cash salary takes a 25% qualifying haircut.
- **S05. Variable Pay / Bonus Share** (+2.0% → Cum. 71.5%): If > 20% of CTC, discounts monthly base income by 15% to buffer bonus volatility.
- **S06. Notice Period / Probation Status** (+2.0% → Cum. 73.5%): Active notice/probation adds 1.50% spread markup and caps principal to 6 months' salary.
- **S07. Liquid Emergency Savings** (+2.5% → Cum. 76.0%): ≥ 6 months reduces required safe cash reserve from 10% to 5%, boosting safe EMI.

- **S08. Credit Card Limit Utilization** (+2.0% → Cum. 78.0%): > 50% utilization adds 0.50% risk spread; < 20% lowers spread by 0.25%.
- **S09. Credit Enquiries (Last 90 Days)** (+1.5% → Cum. 79.5%): > 3 inquiries widens upper rate band by +1.0% (credit hunger signal).
- **S10. Quoted Bank Pre-Approved Rate** (+3.0% → Cum. 82.5%): Benchmarks the exact bank rate on the Negotiation Card to quantify savings.
- **S11. Quoted Processing Fee %** (+2.0% → Cum. 84.5%): Integrated with 18% GST to calculate true all-in APR.
- **S12. Largest Existing Loan Remaining Tenure** (+1.5% → Cum. 86.0%): If < 12 months remaining, projects future cash headroom expansion.
- **S13. Planned Large Outlays (Next 12 Months)** (+1.5% → Cum. 87.5%): Deducts planned one-off capital needs from liquid buffers.
- **S14. Co-Applicant Spouse Salary** (+2.0% → Cum. 89.5%): Combines qualifying income, raising FOIR and safe EMI limits.
- **S15. Home Loan Tax Benefit Status** (+1.5% → Cum. 91.0%): Credits back monthly tax savings (~₹3,000–₹7,000) into disposable cash.
- **S16. EPF / Provident Fund Balance** (+1.5% → Cum. 92.5%): Serves as secondary distress cushion, lowering stress test failure penalty.
- **S17. Comprehensive Health Insurance** (+1.0% → Cum. 93.5%): Lowers required monthly emergency medical cash deduction by 5%.
- **S18. Standalone Unsecured Loan Count** (+1.5% → Cum. 95.0% Max): > 2 active unsecured personal loans adds 1.0% multi-loan leverage penalty.

Self-Employed / MSME Variable Set

- **B01. Documented Annual ITR Profit** (+3.0% → Cum. 63.0%): Hard cap for formal unsecured banking sanction.
- **B02. ITR Filing Vintage** (+2.0% → Cum. 65.0%): ≥ 3 consecutive years of ITR filings cuts base rate spread by 0.75%.
- **B03. Monthly Gross Cash Turnover** (+2.5% → Cum. 67.5%): Evaluates real cash liquidity to set Borrower Safe EMI capacity.
- **B04. Unencumbered Property Ownership** (+3.5% → Cum. 71.0%): **Critical Product Pivot.** If yes, switches loan from unsecured (18%–22%) to LAP (9.5%–11%).
- **B05. Property Market Value** (+2.5% → Cum. 73.5%): Determines maximum LAP sanction capacity using 50%–60% LTV.
- **B06. Title Deed Legal Clarity** (+2.0% → Cum. 75.5%): Clean registered title deed narrows spread; unpartitioned deed widens spread by ±2.0%.
- **B07. Business Operating Vintage** (+2.0% → Cum. 77.5%): ≥ 5 years operating at current location cuts rate spread by 0.50%.
- **B08. Business Premises Ownership** (+1.5% → Cum. 79.0%): Owned shop eliminates rental overhead and increases stability score.
- **B09. Annual GSTR-3B Turnover** (+2.0% → Cum. 81.0%): Unlocks GST surrogate lines (sanctioning up to 10%–15% of verified turnover).
- **B10. Seasonal Cash Inflow Variance** (+1.5% → Cum. 82.5%): If revenue drops > 35% in low season, anchors safe EMI to the trough month.
- **B11. Co-Applicant Family Income** (+2.0% → Cum. 84.5%): Adds verified secondary family cash flow to debt-servicing limits.
- **B12. Commercial Vehicle Fleet Assets** (+1.5% → Cum. 86.0%): Provides secondary movable asset security, reducing equipment rate spreads.
- **B13. Supplier Payment Terms** (+1.5% → Cum. 87.5%): Immediate cash-only suppliers require higher cash cushion, lowering safe EMI.
- **B14. Customer Credit Receivables Days** (+1.5% → Cum. 89.0%): Receivables locked > 45 days applies a 10% haircut on operating cash.
- **B15. Average Daily Bank Balance (ABB)** (+2.0% → Cum. 91.0%): High daily credit balance cuts loan spreads by up to 1.0%.
- **B16. Unsecured / MCA Debt Portion** (+1.5% → Cum. 92.5%): High short-term merchant cash advance debt flags distress, triggering "Borrow Less".
- **B17. Expansion Net Margin Boost** (+1.5% → Cum. 94.0%): Incremental earnings from inventory added to future safe servicing capacity.
- **B18. Statutory Tax / GST Disputes** (+1.0% → Cum. 95.0% Max): Pending statutory notices widens rate spread by 1.50% and bars prime NBFCs.

Informal / Gig Economy Variable Set

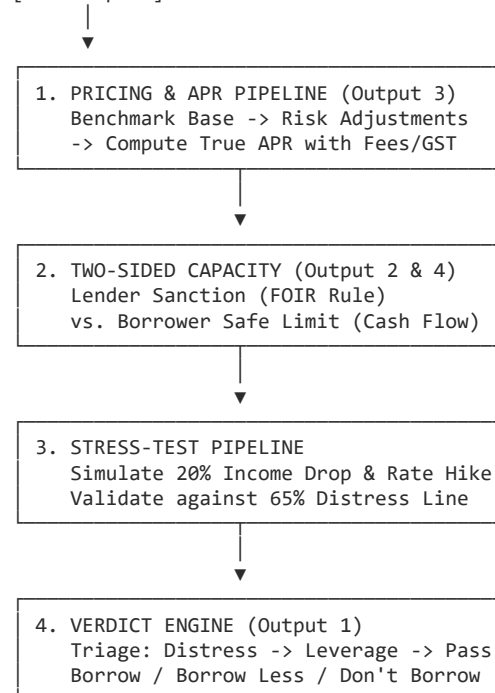
- **I01. Payment Bounces (Last 6 Months)** (+3.5% → Cum. 63.5%): **Kill-switch rule.** Any bounce ≥ 1 forces verdict to **"Don't Borrow"**.
- **I02. Active Instant Digital App Loans** (+2.5% → Cum. 66.0%): Identifies short-term digital leverage; triggers debt consolidation rules.

- **I03. Total App Loan Outstanding Balance** (+2.5% → Cum. 68.5%): Sets mandatory debt clearance requirement before taking new credit.
- **I04. App Loan Interest Rate Level** (+2.5% → Cum. 71.0%): Rates > 24% trigger predatory debt policy, blocking non-essential debt.
- **I05. Productive Equipment Income Boost** (+3.0% → Cum. 74.0%): Incremental income from equipment credited toward future repayment.
- **I06. Payout Mode (Platform vs Cash)** (+2.5% → Cum. 76.5%): Platform digital records unlock priority-sector EV rates (11%–13%) over personal rates.
- **I07. Platform Active Working Days/Month** (+2.0% → Cum. 78.5%): ≥ 24 days validates regular earnings; < 15 days applies 20% volatility haircut.
- **I08. Household Adult Earning Members** (+1.5% → Cum. 80.0%): Quantifies income redundancy to cushion household shocks.
- **I09. Spouse Employment Status** (+2.0% → Cum. 82.0%): Long-term unemployed spouse increases stress test income drop to 30%.
- **I10. Dependent Children Count** (+1.5% → Cum. 83.5%): Adds +₹3,000 per child to non-negotiable living expense floor.
- **I11. Secondary Trade / Craft Skill** (+1.5% → Cum. 85.0%): Secondary informal income cushions platform vehicle downtime.
- **I12. Liquid Emergency Cash / Gold** (+2.0% → Cum. 87.0%): Zero buffer triggers explicit warning: any lost workday causes default.
- **I13. EV Fuel & Maintenance Savings** (+2.0% → Cum. 89.0%): Direct operational savings (~₹3,000–₹5,000/mo) credited toward vehicle EMI.
- **I14. Informal Moneylender Debt** (+1.5% → Cum. 90.5%): Unregistered private debt treated as extreme distress, dropping allowable FOIR to 25%.
- **I15. Commercial Registration / Badge** (+1.5% → Cum. 92.0%): Unlocks asset-hypothecated financing and clean commercial subsidies.
- **I16. Primary Smartphone EMI Status** (+1.0% → Cum. 93.0%): Core tool installment prioritized; default on phone halts all gig earnings.
- **I17. Family Medical Shock (Last 12 Mos)** (+1.0% → Cum. 94.0%): Past uninsured medical debt widens safe capacity uncertainty spread.
- **I18. SHG / JLG Group Membership** (+1.0% → Cum. 95.0% Max): Unlocks priority micro-credit options at capped RBI rates (18%–22%).

3. FORMULAS & VERDICT ENGINE

The underwriting pipeline uses a deterministic mathematical progression to generate the four core outputs:

[User Inputs]



Formula 1: True All-In APR (O_3)

Nominal interest rates hide upfront processing fees and taxes. The all-in APR reflects the true annual internal rate of return (IRR):

1. Upfront Fee Deductions:

$$\text{Total Upfront Fees} = (\text{Loan Principal} \times \text{Processing Fee } \%) \times (1 + \text{GST } 0.18) + \text{Documentation Charges}$$

2. Net Disbursed Principal:

$$\text{Net Disbursed Cash} = \text{Loan Principal} - \text{Total Upfront Fees}$$

3. All-In APR: The annualized rate equating the monthly payments to the net disbursed cash over n periods.

Formula 2: Two-Sided Borrowing Limits (O_2)

- **Lender Maximum Sanction (Bank Formula):**

$$\text{Lender Max EMI} = (\text{Net Monthly Income} \times \text{FOIR } \%) - \text{Existing Monthly EMIs}$$

(FOIR is 50% for Prime Salaried, 45% for MSME, 40% for Informal).

$$\text{Lender Max Principal} = \text{Present Value of (Lender Max EMI, } r_{\text{nominal}}, n)$$

- **Borrower Safe Borrowing Limit (Affordability Formula):**

$$\text{Borrower Safe EMI} = \text{Net Income} - \text{Existing EMIs} - \text{Rent} - \text{Living Expenses} - \text{Emergency Buffer}$$

(Emergency Buffer is 10% of income by default, dropping to 5% if ≥ 6 months liquid savings exist).

$$\text{Borrower Safe Principal} = \text{Present Value of (Borrower Safe EMI, } r_{\text{nominal}}, n)$$

Formula 3: Monthly Payment Ceiling & Stress-Testing (O_4)

- **Baseline Safe Monthly Ceiling:**

$$\text{Monthly EMI Ceiling} = \min(\text{Lender Max EMI, Borrower Safe EMI})$$

- **Stress Test Simulation:** Simulate a 20% drop in net monthly earnings ($I_{\text{stressed}} = I_{\text{net}} \times 0.80$) along with an interest rate hike of +2.0%.

$$\text{Stressed Total Debt Outflow} = \text{Existing EMIs} + \text{Stressed New EMI}$$

- **Safety Condition:**

$$\frac{\text{Stressed Total Debt Outflow}}{I_{\text{stressed}}} \leq 65\%$$

If this ratio exceeds 65%, the safe EMI ceiling is reduced until the test passes.

Formula 4: The Final Verdict Engine (O_1)

Evaluated via a strict decision hierarchy:

1. Kill-Switch Trigger → Verdict: "DON'T BORROW"

- *Condition:* Any payment/auto-debit bounce in the last 6 months ≥ 1 , OR active digital app loans carry interest $> 24\%$ APR, OR safe EMI capacity ≤ 0 .

2. Over-Leverage Trigger → Verdict: "BORROW LESS"

- *Condition:* Requested Loan EMI $>$ Monthly Safe EMI Ceiling, OR Requested Principal $>$ Borrower Safe Principal Limit.

3. Affordability Pass → Verdict: "BORROW"

- *Condition:* Requested Loan EMI $\leq$ Monthly Safe EMI Ceiling, all stress tests pass, and zero kill-switch flags exist.

4. THE TWO-FOLD OUTPUT SYSTEM & "THE WHY" ENGINE

Every single output displayed on the user interface consists of two distinct folds:

- **Fold 1 (The Quantitative Fact):** The exact calculated number or range.
- **Fold 2 (The Qualitative Grounding):** A single plain-English diagnostic sentence explaining the exact limiting factor or policy rule that produced the number.

Fold 1: RESULT VALUE
"Monthly EMI Ceiling: ₹22,000 / month"

Fold 2: THE WHY
"Why: While bank rules allow up to ₹35,000/mo, your ₹28,000 rent and basic living expenses cap safe repayment at ₹22,000 to keep emergency

reserves intact."

Deciding Which "Why" Reason to Display

The engine uses **deterministic priority checks** to select the active reason:

A. Output 1 "Why" Selection (The Verdict)

- **Priority 1 (Distress):** If active bounces exist:

"Why: A loan bounce in the last 6 months triggered our distress safety rule; adding new debt before clearing arrears creates extreme risk of default."

- **Priority 2 (Predatory Debt):** If app loans $> 24\%$ APR exist:

"Why: You carry active digital app debt above 24% APR; taking new loans before consolidating these leads to a compounding debt trap."

- **Priority 3 (Cash Deficit):** If requested EMI $>$ safe EMI:

"Why: Your requested loan requires an EMI of ₹{req_emi}, which overshoots your safe monthly surplus of ₹{safe_emi} by ₹{gap}."

- **Priority 4 (Pass):**

"Why: Your requested monthly EMI of ₹{req_emi} fits within your safe limit of ₹{safe_emi} while keeping your emergency living buffer intact."

B. Output 2 "Why" Selection (Lender Sanction vs. Safe Limit)

- **If Safe Limit $<$ Lender Sanction:**

"Why: Banks will approve up to ₹{lender_max} based purely on gross income, but your rent of ₹{rent} and essential living costs limit your safe debt capacity to ₹{safe_limit}."

- **If Safe Limit $\geq$ Lender Sanction:**

"Why: Your low fixed living expenses and clear debt history allow your safe borrowing limit to match the maximum standard regulatory banking cap."

C. Output 3 "Why" Selection (Fair Rate Band & APR)

- **Priority 1 (Asset Backing):** If property collateral is pledged:

"Why: Pledging an unencumbered shop/property shifts your product to Loan Against Property (LAP), reducing the rate from 18%+ unsecured to {min_rate}%-{max_rate}%."

- **Priority 2 (Prime Credit):** If CIBIL ≥ 750 :

"Why: Your 750+ CIBIL qualifies you for the prime pricing band, while a {fee}% processing fee plus GST sets the true all-in APR at {apr}%."

- **Priority 3 (Unscored/Unknown):** If credit score is unknown:

"Why: A wider rate band of {min_rate}%-{max_rate}% is applied because your credit history is unverified, ensuring transparency without penalizing you as a defaulter."

D. Output 4 "Why" Selection (Safe EMI Ceiling)

- **Priority 1 (Living Expenses Bottleneck):** If Safe EMI $<$ Lender FOIR EMI:

"Why: While bank FOIR norms permit up to ₹{lender_emi}/month, your monthly rent of ₹{rent} and living expenses cap your safe payment at ₹{safe_emi}."

- **Priority 2 (Stress Test Bottleneck):** If safe EMI was lowered to pass income drop:

"Why: Under a 20% income downturn, an EMI above ₹{safe_emi} would consume over 65% of your remaining cash flow, so the ceiling was adjusted downward."

- **Priority 3 (Bank Regulatory Cap):** If FOIR is the tighter constraint:

"Why: Capped at ₹{safe_emi} because regulatory guidelines prevent total monthly debt repayments from exceeding 50% of your take-home pay."

5. THE NEGOTIATION CARD ENGINE

The Negotiation Card is a mobile-optimized summary the borrower can reference during a discussion with a loan officer. It transforms output figures into clear counter-arguments.

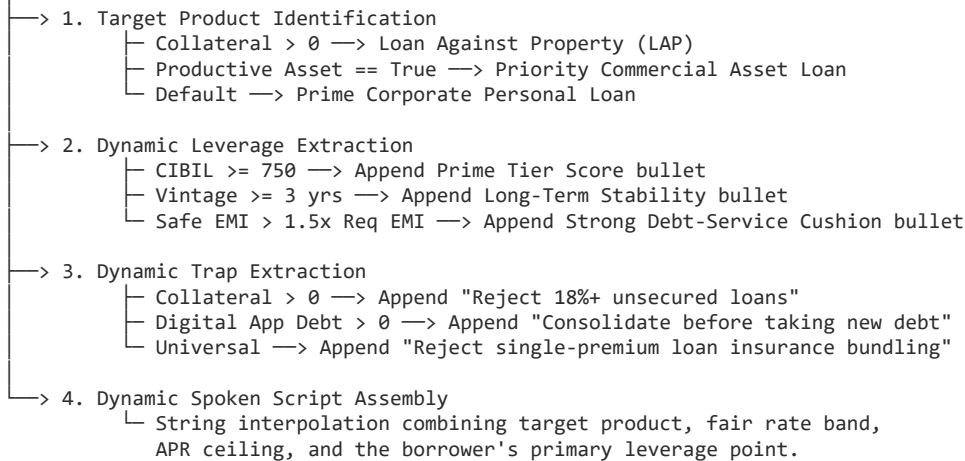
The Universal Card Structure

Regardless of the borrower's background, the card dynamically generates four zones:

1. **Target Terms (Anchors):** Fair interest rate band, hard APR cap, and maximum acceptable processing fee.
2. **Borrower Leverage Points:** Positive data points extracted from their inputs (e.g., CIBIL score, low FOIR, collateral, or vintage).
3. **Bank Traps to Reject:** Specific terms to decline (e.g., bundled credit insurance, express unsecured loans, or inflated valuation charges).
4. **Verbatim Script:** A word-for-word counter-script the user can read directly to the bank manager.

Generation Logic

Profile Inputs + Output Engine Calculations



Dynamic Script Assembly Template

The spoken script is assembled using this dynamic string structure:

"I am applying for a {Target Product}. Based on {Primary Leverage Point}, standard regulatory pricing indicates a fair rate of {Min Rate}% to {Max Rate}%, with an all-in APR cap of {All-In APR}%. My repayment ceiling is strictly ₹{Safe Monthly EMI}/month; I will not accept structured tenures, fee markups above {Fee Cap}, or mandatory single-premium insurance add-ons that breach these limits."

6. IMPLEMENTATION & RUNTIME INSTRUCTIONS

When implementing this system in code:

1. **Zero External Dependencies:** Use pure Python standard libraries and math functions; do not connect to external APIs or persistent databases.
2. **State & Confidence:** Initialize user session state with confidence = 60.0% upon completing mandatory questions, scaling up to 95.0% as optional inputs are filled.
3. **Output Bundling:** Store outputs as dictionary objects containing both the numeric value and the "why" string to ensure they render together in the UI.
4. **Clean UI Separation:** Keep underwriting equations in a dedicated logic module (e.g., underwriting.py) and user interface rendering in a separate presentation layer.

User prompt: create a readme file where all of this will be present

Response: # Deterministic Borrowing Underwriting Engine & Copilot

An explainable, deterministic financial underwriting engine designed for the Indian retail lending landscape. The en

1. Categories & Persona Archetypes

The copilot evaluates applicants across three distinct categories, reflecting the structural realities of Indian ret

Salaried (Prime Corporate / Institutional)

* **Archetype Example:** Priya (Software Engineer, Bengaluru)

* **Underwriting Focus:** Prime pricing optimization, processing fee/GST reduction, and evaluating discretionary cas

* **Failure Mode:** High lifestyle overhead (rent, credit card spending) reducing real disposable income despite a h

* **Target Product:** Unsecured Prime Corporate Personal Loan.

Self-Employed / MSME (Merchants & Small Enterprises)

* **Archetype Example:** Ravi (Kirana Store Owner, Mysuru)

* **Underwriting Focus:** Collateral discovery and product pivoting (diverting applicants from predatory unsecured b

* **Failure Mode:** Conservative ITR filings leading traditional algorithms to reject or penalize the borrower with
 * **Target Product:** Loan Against Property (LAP) / Secured Mortgage Line.

Informal / Gig Economy (Platform & Daily Earners)

* **Archetype Example:** Anita (Delivery Fleet Partner, Hubballi)
 * **Underwriting Focus:** Distress screening (kill-switches on active bounces or high-cost digital app debt) and val
 * **Failure Mode:** Stacking daily micro-loans and digital payday credit apps (>30% APR) for non-productive consumpt
 * **Target Product:** Priority Sector Commercial EV / Productive Asset Hypothecation.

2. Two-Tier Question Framework & Confidence Engine

The engine adheres to a core underwriting principle: ****Confidence widens with silence; information narrows bands****.

* **Baseline (60% Confidence):** Answering the 10 Universal Mandatory questions provides an immediate preliminary as
 * **Scaling to 95% Confidence:** Each category-specific variable question answered adds a calibrated percentage incr
 * **95% Ceiling:** The confidence meter caps at 95% to acknowledge that self-reported borrower data lacks direct bur

[10 Universal Mandatory Questions] —> 60.0% Base Confidence (Wide Output Bands) | ▼ [Adaptive Variable Questions (1-18)] —>
 +1.0% to +3.5% per answered question | ▼ [Full Category Completion] —> 95.0% Max Confidence (Tight Output Bands)

Universal Mandatory Questions (10 Questions – Base 60%)

ID	Parameter	Input Type	Metric Impacted	Underwriting Mechanics
---	---	---	---	---
M01	Loan Purpose	Select	01, 03	Sets product classification and default tenure limits.
M02	Target Loan Amount	Numeric (₹)	01, 02	Sets requested principal to evaluate against safe borrowing
M03	Preferred Tenure	Months	04	Defines amortization schedule (\$n\$).
M04	Employment Structure	Select	Branching	Sets regulatory FOIR benchmark (40%-55%) and activates categ
M05	Net Monthly Take-Home	Numeric (₹/mo)	02, 04	Baseline denominator for all debt-to-income and cash f
M06	Existing Monthly EMIs	Numeric (₹/mo)	02, 04	Directly deducted from allowable debt headroom.
M07	Monthly Rent / Housing	Numeric (₹/mo)	02, 04	Subtracted in borrower residual cash flow calculation
M08	Essential Living Costs	Numeric (₹/mo)	02, 04	Non-negotiable living expense floor.
M09	Credit Score (CIBIL)	Select	03	Adjusts base pricing; unverified score widens bands without default
M10	Borrower Age	Numeric (Yrs)	02, 04	Caps loan tenure based on a standard 60-year retirement boundary

Variable Questions & Confidence Allocation (18 Per Category)

Salaried Variable Set

* **S01. Employer Corporate Classification** (+3.0% \$→\$ 63.0%): Tier-1/Govt raises FOIR to 55% and cuts sp
 * **S02. Current Job Vintage** (+2.5% \$→\$ 65.5%): \$ge 3\$ years narrows rate spread by \$pm 0.50\%\$; \$<1\$
 * **S03. Total Career Experience** (+2.0% \$→\$ 67.5%): \$>5\$ years unlocks maximum 60-72 month tenures; \$<2\$
 * **S04. Salary Credit Method** (+2.0% \$→\$ 69.5%): Direct bank transfer validates income at 100%; cash sal
 * **S05. Variable Pay / Bonus Share** (+2.0% \$→\$ 71.5%): If \$>20\%\$, discounts monthly base income by 15%
 * **S06. Notice Period / Probation Status** (+2.0% \$→\$ 73.5%): Active notice/probation adds 1.50% spread m
 * **S07. Liquid Emergency Savings** (+2.5% \$→\$ 76.0%): \$ge 6\$ months reduces required cash reserve from 1
 * **S08. Credit Card Limit Utilization** (+2.0% \$→\$ 78.0%): \$>50\%\$ utilization adds 0.50% spread penalty;
 * **S09. Bureau Inquiries (Last 90 Days)** (+1.5% \$→\$ 79.5%): \$>3\$ inquiries widens upper rate band by +1.
 * **S10. Quoted Bank Pre-Approved Rate** (+3.0% \$→\$ 82.5%): Replaces generic baseline with actual quoted r
 * **S11. Quoted Processing Fee** (+2.0% \$→\$ 84.5%): Factored into internal rate formulas with 18% GST to
 * **S12. Existing Loan Remaining Tenure** (+1.5% \$→\$ 86.0%): If \$<12\$ months remaining, projects future st
 * **S13. Planned Large Outlays (12 Mos)** (+1.5% \$→\$ 87.5%): Deducts planned capital expenditures from liq
 * **S14. Co-Applicant Net Income** (+2.0% \$→\$ 89.5%): Combines with primary income to increase lender FOIR
 * **S15. Home Loan Tax Benefit Status** (+1.5% \$→\$ 91.0%): Credits back monthly tax savings (~₹3,000-₹7,00
 * **S16. EPF / PF Balance Available** (+1.5% \$→\$ 92.5%): Serves as secondary distress cushion, lowering st
 * **S17. Comprehensive Health Insurance** (+1.0% \$→\$ 93.5%): Lowers required monthly emergency medical cas
 * **S18. Standalone Unsecured Loan Count** (+1.5% \$→\$ 95.0% Max): \$>2\$ active unsecured personal loans add

Self-Employed / MSME Variable Set

* **B01. Documented Annual ITR Profit** (+3.0% \$→\$ 63.0%): Hard cap for formal unsecured banking sanction.
 * **B02. ITR Filing Vintage** (+2.0% \$→\$ 65.0%): \$ge 3\$ consecutive years of ITR filings cuts base rate s
 * **B03. Monthly Gross Cash Intake** (+2.5% \$→\$ 67.5%): Evaluates actual operational cash flow to establis
 * **B04. Unencumbered Property Ownership** (+3.5% \$→\$ 71.0%): ****Critical Product Pivot.**** Switches loan fr
 * **B05. Property Market Value** (+2.5% \$→\$ 73.5%): Computes maximum LAP sanction capacity using 50%-60% L
 * **B06. Title Deed Legal Clarity** (+2.0% \$→\$ 75.5%): Clean registered title deed narrows spread; unparti
 * **B07. Business Operating Vintage** (+2.0% \$→\$ 77.5%): \$ge 5\$ years operating at current location cuts
 * **B08. Business Premises Ownership** (+1.5% \$→\$ 79.0%): Owned shop eliminates rental overhead and raises
 * **B09. Annual GSTR-3B Turnover** (+2.0% \$→\$ 81.0%): Unlocks GST surrogate lines (sanctioning up to 10%-1
 * **B10. Seasonal Cash Inflow Variance** (+1.5% \$→\$ 82.5%): If revenue drops \$>35\%\$ in low season, anchor
 * **B11. Co-Applicant Family Income** (+2.0% \$→\$ 84.5%): Adds verified secondary family cash flow to debt-
 * **B12. Commercial Vehicle Fleet Assets** (+1.5% \$→\$ 86.0%): Provides secondary movable asset security, r
 * **B13. Supplier Payment Terms** (+1.5% \$→\$ 87.5%): Immediate cash-only suppliers require higher cash cus
 * **B14. Customer Credit Receivables Days** (+1.5% \$→\$ 89.0%): Receivables locked \$>45\$ days applies a 10%
 * **B15. Average Daily Bank Balance (ABB)** (+2.0% \$→\$ 91.0%): High daily credit balance cuts loan spreads
 * **B16. Unsecured / MCA Debt Portion** (+1.5% \$→\$ 92.5%): High short-term merchant cash advance debt flag

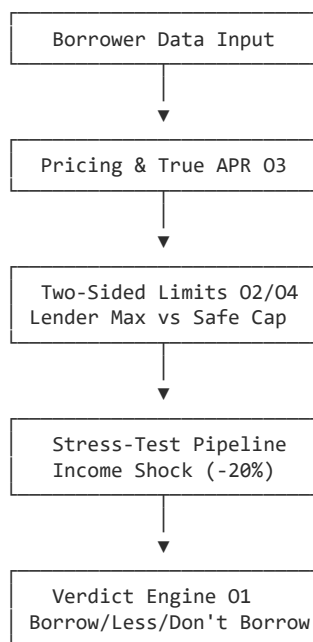
* **B17. Expansion Net Margin Boost** (+1.5% $\rightarrow$ 94.0%): Incremental earnings from inventory added to futu
 * **B18. Statutory Tax / GST Disputes** (+1.0% $\rightarrow$ 95.0% Max): Pending statutory notices widens rate sprea

Informal / Gig Economy Variable Set

* **I01. Payment Bounces (Last 6 Months)** (+3.5% $\rightarrow$ 63.5%): **Kill-switch rule.** Any bounce $\geq$ 1\$ for
 * **I02. Active Instant Digital App Loans** (+2.5% $\rightarrow$ 66.0%): Identifies short-term digital leverage; tri
 * **I03. Total App Loan Outstanding Balance** (+2.5% $\rightarrow$ 68.5%): Sets mandatory debt clearance requirement
 * **I04. App Loan Interest Rate Level** (+2.5% $\rightarrow$ 71.0%): Rates $>24\%$ trigger predatory debt policy, bl
 * **I05. Productive Equipment Income Boost** (+3.0% $\rightarrow$ 74.0%): Incremental income from equipment credited
 * **I06. Payout Mode (Platform vs Cash)** (+2.5% $\rightarrow$ 76.5%): Platform digital records unlock priority-sect
 * **I07. Platform Active Working Days/Month** (+2.0% $\rightarrow$ 78.5%): ≥ 24 days validates regular earnings;
 * **I08. Household Adult Earning Members** (+1.5% $\rightarrow$ 80.0%): Quantifies income redundancy to cushion hous
 * **I09. Spouse Employment Status** (+2.0% $\rightarrow$ 82.0%): Long-term unemployed spouse increases stress test i
 * **I10. Dependent Children Count** (+1.5% $\rightarrow$ 83.5%): Adds + $\text{₹}3,000$ per child to non-negotiable living exp
 * **I11. Secondary Trade / Craft Skill** (+1.5% $\rightarrow$ 85.0%): Secondary informal income cushions platform ve
 * **I12. Liquid Emergency Cash / Gold** (+2.0% $\rightarrow$ 87.0%): Zero buffer triggers explicit warning: any lost
 * **I13. EV Fuel & Maintenance Savings** (+2.0% $\rightarrow$ 89.0%): Direct operational savings ($\sim \text{₹}3,000\text{--}\text{₹}5,000/\text{mo}$)
 * **I14. Informal Moneylender Debt** (+1.5% $\rightarrow$ 90.5%): Unregistered private debt treated as extreme distr
 * **I15. Commercial Registration / Badge** (+1.5% $\rightarrow$ 92.0%): Unlocks asset-hypothecated financing and cle
 * **I16. Primary Smartphone EMI Status** (+1.0% $\rightarrow$ 93.0%): Core tool installment prioritized; default on
 * **I17. Family Medical Shock (Last 12 Mos)** (+1.0% $\rightarrow$ 94.0%): Past uninsured medical debt widens safe c
 * **I18. SHG / JLG Group Membership** (+1.0% $\rightarrow$ 95.0% Max): Unlocks priority micro-credit options at capp

3. Mathematical Formulas & Verdict Engine

The underwriting pipeline computes four deterministic outputs ($\$0_1$ to $\$0_4$):



1. Output 3 ($\$0_3$): Fair Interest Rate & All-In APR

* **Nominal Rate Band:** Starts at the product benchmark base, discounted by positive credentials (high CIBIL, verif
 * **True All-In APR Calculation:**

$$\text{\text{\text{Upfront Fees}}} = (\text{\text{Principal}} \times \text{\text{Processing Fee \%}}) \times (1 + \text{\text{GST}} \times 0.18) + \text{\text{D}}$$

$$\text{\text{Net Disbursed Cash}} = \text{\text{Principal}} - \text{\text{Upfront Fees}}$$

$$\text{\text{APR}} = \text{\text{Internal Rate of Return equating monthly EMIs over } n \text{ periods to Net Disbursed Cash}}$$

2. Output 2 ($\$0_2$): Two-Sided Borrowing Limits

* **Lender Maximum Sanction (Bank Formula):**

$$\text{\text{Lender Max EMI}} = (\text{\text{Net Income}} \times \text{\text{FOIR \%}}) - \text{\text{Existing EMIs}}$$

$$\text{\text{Lender Sanction Limit}} = \text{\text{Present Value}}(\text{\text{Rate}}, n, \text{\text{Lender Max EMI}})$$
 (FOIR benchmark: Salaried Prime = 50%-55%, MSME = 45%, Informal = 40%).
 * **Borrower Safe Borrowing Limit (Affordability Formula):**

$$\text{\text{Borrower Safe EMI}} = \text{\text{Net Income}} - \text{\text{Existing EMIs}} - \text{\text{Rent}} - \text{\text{Living Expenses}} - \text{\text{te}}$$

$$\text{\text{Borrower Safe Limit}} = \text{\text{Present Value}}(\text{\text{Rate}}, n, \text{\text{Borrower Safe EMI}})$$
 (Default emergency buffer = 10% of net income; drops to 5% if ≥ 6 months liquid runway exists).

3. Output 4 ($\$0_4$): Safe Monthly Payment Ceiling & Stress Resilience

* **Safe Ceiling Value:**

$$\text{\text{Monthly EMI Ceiling}} = \min(\text{\text{Lender Max EMI}}, \text{\text{Borrower Safe EMI}})$$
 * **Shock Simulation:**
 Simulate a 20% drop in take-home income along with a $+2.0\%$ rate spike on floating debt:

$$\text{Stressed Total Debt Outflow} = \text{Existing EMIs} + \text{Stressed New EMI}$$

$$\text{Distress Ratio} = \frac{\text{Stressed Total Debt Outflow}}{\text{Net Income}} \times 0.80$$
 If $\text{Distress Ratio} > 65\%$, the ceiling is iteratively reduced until the repayment survives the simulated s

4. Output 1 (\$0_1\$): Final Recommendation Verdict

Evaluated using a strict three-tier triage hierarchy:

- Kill-Switch Trigger** $\rightarrow$ Verdict: "DON'T BORROW"
 - Condition: Any auto-debit bounce in the last 6 months $\geq 1\%$, OR active digital app debt $> 24\%$ APR, OR safe
- Capacity Over-Leverage** $\rightarrow$ Verdict: "BORROW LESS"
 - Condition: Requested EMI > Monthly Safe EMI Ceiling, OR Requested Principal > Borrower Safe Limit.
- Affordability Pass** $\rightarrow$ Verdict: "BORROW"
 - Condition: Requested EMI $\leq$ Monthly Safe EMI Ceiling, zero kill-switch triggers, and the repayment schedule

4. The Two-Fold Output System & "The Why" Engine

Every output is rendered as a **two-folded card**:

- Fold 1:** The hard quantitative value (e.g., $\text{₹}22,000/\text{mo}$, $10.5\% - 11.25\%$, or **Borrow Less**).
- Fold 2:** A plain-English diagnostic sentence explaining the exact limiting factor or policy rule that produced

QUANTITATIVE FACT	Monthly EMI Ceiling: ₹22,000 / month	Fold 1:
GROUNDING (THE WHY)	"Why: While bank rules allow up to ₹35,000/mo, your ₹28,000 rent and basic living expenses cap safe repayment at ₹22,000 to keep emergency reserves intact."	Fold 2: QUALITATIVE

Deterministic Bottleneck Logic for Selecting "The Why"

Output 1: The Verdict Explanation

- Distress Priority:** If bounces $\geq 1\%$:
 - Why: A loan bounce in the last 6 months triggered our distress safety rule; adding new debt before clearing arre
- Predatory Debt Priority:** If app loans $> 24\%$ APR:
 - Why: You carry active digital app debt above 24% APR; taking new loans before consolidating these leads to a com
- Capacity Deficit Priority:** If Requested EMI > Safe EMI:
 - Why: Your requested loan requires an EMI of $\text{₹}\{\text{req_emi}\}$, which overshoots your safe monthly surplus of $\text{₹}\{\text{safe_emi}\}$
- Affordability Pass:**
 - Why: Your requested monthly EMI of $\text{₹}\{\text{req_emi}\}$ fits within your safe limit of $\text{₹}\{\text{safe_emi}\}$ while keeping your emer

Output 2: Maximum Amount Gap Explanation

- If Safe Limit < Lender Sanction:**
 - Why: Banks will approve up to $\text{₹}\{\text{lender_max}\}$ based purely on gross income, but your rent of $\text{₹}\{\text{rent}\}$ and essential
- If Safe Limit $\geq$ Lender Sanction:**
 - Why: Your low fixed living expenses and clear debt history allow your safe borrowing limit to match the maximum

Output 3: Fair Rate Band & APR Explanation

- Asset Collateral Priority:** If property pledged:
 - Why: Pledging an unencumbered shop/property shifts your product to Loan Against Property (LAP), reducing the rat
- Prime Bureau Priority:** If CIBIL ≥ 750 :
 - Why: Your 750+ CIBIL qualifies you for the prime pricing band, while a $\{\text{fee}\}\%$ processing fee plus GST sets the t
- Unscored/Unknown Priority:** If CIBIL unverified:
 - Why: A wider rate band of $\{\text{min_rate}\}\% - \{\text{max_rate}\}\%$ is applied because your credit history is unverified, ensuring

Output 4: Safe Monthly EMI Ceiling Explanation

- Cash Flow Priority:** If Safe EMI < Lender FOIR EMI:
 - Why: While bank FOIR norms permit up to $\text{₹}\{\text{lender_emi}\}/\text{month}$, your monthly rent of $\text{₹}\{\text{rent}\}$ and living expenses ca
- Stress Shock Priority:** If ceiling was adjusted downward during shock simulation:
 - Why: Under a 20% income downturn, an EMI above $\text{₹}\{\text{safe_emi}\}$ would consume over 65% of your remaining cash flow, s
- Regulatory Cap Priority:** If FOIR is the tighter bound:
 - Why: Capped at $\text{₹}\{\text{safe_emi}\}$ because regulatory guidelines prevent total monthly debt repayments from exceeding 50

5. Tactical Negotiation Card Engine

The Negotiation Card converts backend calculations into actionable counter-benchmarks that the borrower can referenc

MOBILE NEGOTIATION CARD	
Product: Prime Unsecured Corporate Loan	1. TARGET ANCHORS Target
• Fair Rate Band: 10.50% – 11.25% • Hard APR Cap: 11.80% (Processing Fee $\leq 1.0\%$ + GST) • Safe EMI Limit: ₹22,000 / month	3. BANK TRAPS
• CIBIL score of 780 qualifies for prime corporate tier. • Tier-1 listed corporate employer salary account. • Existing FOIR < 20% leaves ample debt capacity.	2. BORROWER LEVERAGE POINTS
TO REJECT • Mandatory single-premium credit shield insurance. • Processing fees exceeding 1.0% + GST. • Prepayment lock-ins exceeding 12 months.	4. VERBATIM
SPOKEN COUNTER-SCRIPT "My CIBIL is 780 and my employer is Category-A. Standard corporate pricing clears between 10.5% and	

11.25%. If | | the APR exceeds 11.80% with processing fees capped at | | 1%, I will move the sanction to my primary salary bank. | | Do not deduct loan insurance from disbursement." |

Dynamic Assembly Logic

* **Target Product Identification:** Evaluates collateral assets and equipment utility flags to assign the appropriate
* **Leverage Extraction:** Surfaces verified strengths (e.g., CIBIL ≥ 750 , stable operational vintage ≥ 3 years)
* **Trap Identification:** Identifies common retail lending markups based on the borrower's profile (e.g., bundled services)
* **Spoken Script Templating:** Compiles targets and leverage points into a conversational counter-statement:
> "I am applying for a **{Target Product}**. Based on **{Primary Leverage Point}**, standard regulatory pricing is

6. Implementation Guidelines

* **Zero External Dependencies:** Core underwriting calculations must remain entirely self-contained within native Python
* **State Management:** Session states initialize at `confidence = 60.0%` upon answering mandatory inputs, scaling incrementally
* **Paired Data Bundles:** Ensure the calculation layer returns both the numeric metric and its corresponding diagnostic
* **Separation of Concerns:** Keep presentation components strictly decoupled from the mathematical engine, ensuring

User prompt: give every info regarding the project so far in one single file only that too very descriptively, I will give this to antigravity agent to build the app

Response: # DETERMINISTIC BORROWING UNDERWRITING COPILOT

Complete System Architecture, Financial Rules, and Agent Implementation Specification

This single master document contains the complete specifications, domain rules, calculation formulas, confidence protocols,

1. SYSTEM ARCHITECTURE & CORE DESIGN PRINCIPLES

1.1 Architecture & Separation of Concerns

The application is structured into two completely decoupled layers:

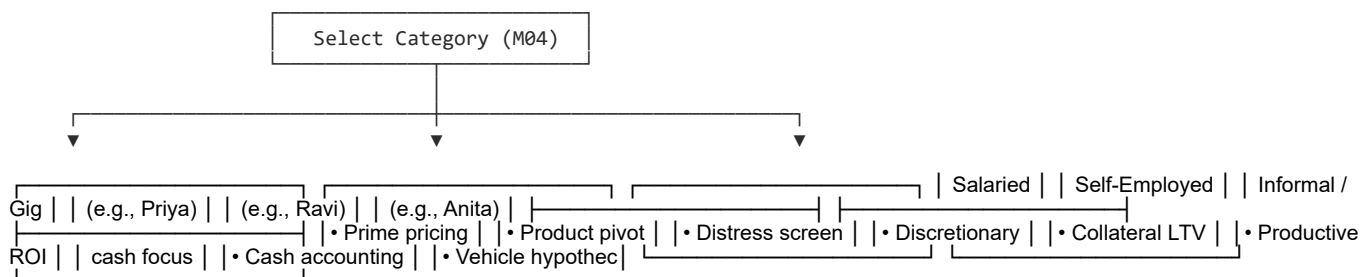
- * **The Underwriting Engine** (`engine/underwriting.py`): A pure, deterministic Python module executing mathematical logic
- * **The Presentation Layer** (`app.py` / UI): A mobile-first interactive interface (Streamlit, Pure HTML/JS, or Flutter)

1.2 Core Evaluation Principles

- * **Rule 1: Deterministic Transparency:** Every calculation is traceable to mathematical formulas and codified Indian regulations
- * **Rule 2: Confidence Widens with Silence:** Fewer answers produce wider output ranges and lower confidence scores; more inputs increase precision
- * **Rule 3: Unknown is Never Zero:** If a borrower enters "Unknown" (e.g., credit score or property valuation), the system flags for manual review
- * **Rule 4: Every Number Has a Why:** Every numerical output ($\$0.1$ to $\$0.4$) is paired directly with an explicit, actionable reason

2. BORROWER CATEGORIES & PERSONA SPECIFICATIONS

The engine customizes underwriting logic across three primary borrower classes:



2.1 Salaried (Prime Corporate / Institutional)

* **Target Archetype:** Priya (29, Software Engineer in Bengaluru; ₹1,40,000 net take-home; 780 CIBIL; seeking ₹8,00,000)
* **Underwriting Focus:** Securing prime personal loan pricing (10.5%–11.5%), capping processing fee drag, and assessing creditworthiness
* **Failure Mode:** Banks approve high loan amounts based strictly on gross salary, while heavy fixed lifestyle overruns
* **Target Product:** Unsecured Prime Corporate Personal Loan.

2.2 Self-Employed / MSME (Merchants & Small Enterprises)

* **Archetype Example:** Ravi (38, Kirana Store Owner in Mysuru; ₹85,000 monthly cash intake; low reported ITR profile)
* **Underwriting Focus:** Product Pivoting. Diverting thin-ITR or cash-heavy merchants away from predatory unsecured loans
* **Failure Mode:** Thin formal tax returns trigger automated rejections from prime bank algorithms, driving the borrower to informal lenders
* **Target Product:** Secured Loan Against Property (LAP) / Micro-Mortgage.

2.3 Informal / Gig Economy (Platform & Daily Earners)

* **Archetype Example:** Anita (32, Platform Delivery Fleet Partner in Hubballi; ₹28,000 net monthly earnings; 1 recent order)

* **Underwriting Focus:** Distress Screening & Productive Asset ROI. Activating a mandatory kill-switch if recent
 * **Failure Mode:** Rolling over payday app loans charging >30% APR; borrowing for non-productive consumption.
 * **Target Product:** Priority Sector Commercial EV / Productive Asset Hypothecation.

3. MASTER QUESTION MATRIX & LIVE CONFIDENCE PROGRESSION (104 ROWS)

The live confidence bar begins at **0%**, scales to **60.0%** upon completing the 10 Universal Mandatory questions,

3.1 104-Row Excel Lookup Structure

*(Rows 1-10: Mandatory; Rows 11-28: Salaried Tier-1; Rows 29-46: Salaried SME; Rows 47-64: Salaried Startup; Rows 65

Row	Question ID	Tier	Category Target	Parameter & Question Summary	Input Type	Added Conf %	Cumulative
1	**M01**	Mandatory	Universal	Primary loan purpose Select 6.00% 6.00% Sets baseline product risk c			
2	**M02**	Mandatory	Universal	Target requested loan principal (₹) Numeric 6.00% 12.00% Requested p			
3	**M03**	Mandatory	Universal	Target repayment tenure (Months) Numeric 6.00% 18.00% Sets amortizat			
4	**M04**	Mandatory	Universal	Primary employment structure Select 6.00% 24.00% Sets regulatory FOI			
5	**M05**	Mandatory	Universal	Net monthly in-hand income (₹) Numeric 6.00% 30.00% Income denominat			
6	**M06**	Mandatory	Universal	Total existing monthly loan EMIs (₹) Numeric 6.00% 36.00% Existing d			
7	**M07**	Mandatory	Universal	Monthly rent / housing outflow (₹) Numeric 6.00% 42.00% Shelter cost			
8	**M08**	Mandatory	Universal	Essential household living costs (₹) Numeric 6.00% 48.00% Basic livi			
9	**M09**	Mandatory	Universal	Credit score (CIBIL) Select 6.00% 54.00% Adjusts base pricing; unver			
10	**M10**	Mandatory	Universal	Borrower age (Years) Numeric 6.00% **60.00%** Caps maximum tenure a			
11	**S01**	Variable	Salaried	Employer corporate tier Select 3.00% 63.00% Tier-1/Govt raises FOIR t			
12	**S02**	Variable	Salaried	Current job vintage (Years) Numeric 2.50% 65.50% \$ge 3\$ years narrow			
13	**S03**	Variable	Salaried	Total career experience (Years) Numeric 2.00% 67.50% \$>5\$ years unloc			
14	**S04**	Variable	Salaried	Salary payment mode (Bank vs Cash) Select 2.00% 69.50% Bank transfer			
15	**S05**	Variable	Salaried	Variable bonus share of CTC (%) Numeric 2.00% 71.50% If \$>20\%\$, disc			
16	**S06**	Variable	Salaried	Notice period / probation status Select 2.00% 73.50% Active notice/pr			
17	**S07**	Variable	Salaried	Liquid emergency runway (Months) Numeric 2.50% 76.00% \$ge 6\$ mos cut			
18	**S08**	Variable	Salaried	Credit card limit utilization (%) Numeric 2.00% 78.00% \$>50\%\$ adds 0			
19	**S09**	Variable	Salaried	Credit inquiries in last 90 days Numeric 1.50% 79.50% \$>3\$ inquiries			
20	**S10**	Variable	Salaried	Pre-approved bank quote rate (%) Numeric 3.00% 82.50% Sets benchmark			
21	**S11**	Variable	Salaried	Quoted processing fee (%) Numeric 2.00% 84.50% Factored into internal			
22	**S12**	Variable	Salaried	Remaining months on largest debt Numeric 1.50% 86.00% If \$<12\$ mos re			
23	**S13**	Variable	Salaried	Planned large expenses (Next 12 mos) Select 1.50% 87.50% Deducts plan			
24	**S14**	Variable	Salaried	Contributing co-applicant net income Numeric 2.00% 89.50% Combines qu			
25	**S15**	Variable	Salaried	Home loan tax rebate utilization Select 1.50% 91.00% Credits back mon			
26	**S16**	Variable	Salaried	EPF balance above ₹3,00,000 Select 1.50% 92.50% Distress backstop ass			
27	**S17**	Variable	Salaried	Comprehensive health insurance Select 1.00% 93.50% Lowers required mo			
28	**S18**	Variable	Salaried	Unsecured personal loan count Numeric 1.50% **95.00%** \$>2\$ active un			
29-46	-	Variable	Salaried (SME)	S01-S18 Repeated for SME Category Various - Scaled 63%-95% Identic			
47-64	-	Variable	Salaried (Startup)	S01-S18 Repeated for Startup Category Various - Scaled 63%-95%			
65-82	-	Variable	Salaried (PSU)	S01-S18 Repeated for PSU/Govt Category Various - Scaled 63%-95% Id			
83	**B01**	Variable	Self-Employed	Net taxable profit on latest ITR (₹) Numeric 3.00% 63.00% Hard c			
84	**B02**	Variable	Self-Employed	Consecutive ITR filing years Numeric 2.50% 65.50% \$ge 3\$ consec			
85	**B03**	Variable	Self-Employed	Gross monthly cash intake (₹) Numeric 2.50% 68.00% Evaluates act			
86	**B04**	Variable	Self-Employed	Unencumbered property ownership Select 3.50% 71.50% **Product Pi			
87	**B05**	Variable	Self-Employed	Property conservative market value (₹) Numeric 3.00% 74.50% Comp			
88	**B06**	Variable	Self-Employed	Property registered title deed clarity Select 2.50% 77.00% Clean			
89	**B07**	Variable	Self-Employed	Business vintage at location (Years) Numeric 3.00% 80.00% \$ge 5			
90	**B08**	Variable	Self-Employed	Business premises ownership type Select 3.00% 83.00% Owned premi			
91	**B09**	Variable	Self-Employed	Annualized GSTR-3B turnover (₹) Numeric 4.00% 87.00% Enables GST			
92	**B10**	Variable	Self-Employed	Seasonal revenue variance (>35%) Select 4.00% 91.00% High season			
93	**B11**	Variable	Self-Employed	Earning spouse / co-applicant income Numeric 4.00% **95.00%** Ad			
94	**I01**	Variable	Informal / Gig	Auto-debit / EMI bounces in last 6 mos Select 3.50% 63.50% **Ki			
95	**I02**	Variable	Informal / Gig	Active instant digital app loan count Numeric 3.00% 66.50% Iden			
96	**I03**	Variable	Informal / Gig	Outstanding digital app debt (₹) Numeric 3.00% 69.50% Quantifie			
97	**I04**	Variable	Informal / Gig	App loan interest rates (>24% APR) Select 3.00% 72.50% Rates \$>			
98	**I05**	Variable	Informal / Gig	Monthly income boost from asset (₹) Numeric 3.50% 76.00% Increm			
99	**I06**	Variable	Informal / Gig	Payout mode (Digital app vs Cash) Select 3.00% 79.00% Digital r			
100	**I07**	Variable	Informal / Gig	Platform active delivery days/month Numeric 3.00% 82.00% \$ge			
101	**I08**	Variable	Informal / Gig	Liquid emergency cash or gold (₹) Numeric 3.00% 85.00% Zero bu			
102	**I09**	Variable	Informal / Gig	EV monthly fuel/maintenance savings Numeric 3.50% 88.50% Direc			
103	**I10**	Variable	Informal / Gig	Informal neighborhood debt status Select 3.50% 92.00% Unregist			
104	**I11**	Variable	Informal / Gig	Registered SHG / JLG membership Select 3.00% **95.00%** Active			

4. MATHEMATICAL FORMULAS & CALCULATION ENGINE

[Borrower Inputs] —> [1. Rate & True APR O3] —> [2. Two-Sided Limits O2/O4] —> [3. Stress Test] —> [4. Verdict Engine O1]

4.1 Output 3 (\$0_3\$): Fair Interest Rate Band & All-In APR

1. **Base Rate Lookup:**

* Unsecured Prime Salaried: \$11.0\% - 13.5\%\$

```

Check Tier 3: Affordability & Stress Clearance
└─ Requested Loan EMI <= Monthly Safe EMI Ceiling AND Stress Ratio <= 65%
──> Verdict = "BORROW"

```

Deterministic Bottleneck Logic for Selecting "The Why"

Output 1 "Why" (The Verdict)

- Trigger 1 (Bounce Kill-Switch): `bounces_last_6m >= 1`

"Why: An auto-debit bounce in the last 6 months flags high default distress; adding new debt before clearing arrears creates severe insolvency risk."
- Trigger 2 (Predatory Debt Trap): `has_high_cost_app_debt == True`

"Why: You carry active digital app debt above 24% APR; borrowing more before consolidating these loans deepens a compounding debt cycle."
- Trigger 3 (Payment Overshoots Surplus): `requested_emi > safe_emi_ceiling`

"Why: Your requested loan requires an EMI of ₹{req_emi:,}, which overshoots your safe monthly surplus of ₹{safe_emi:,} by ₹{gap:,}."
- Trigger 4 (Clean Approval): `verdict == "BORROW"`

"Why: Your requested monthly EMI of ₹{req_emi:,} fits within your safe limit of ₹{safe_emi:,} while keeping your emergency living buffer intact."

Output 2 "Why" (Lender Sanction vs. Safe Limit)

- Trigger 1 (Living Costs Bound Borrowing): `safe_limit < lender_sanction`

"Why: Banks will approve up to ₹{lender_sanction:,} based purely on gross income, but your fixed rent of ₹{rent:,} and essential living costs limit your safe debt capacity to ₹{safe_limit:,}."
- Trigger 2 (Low Living Costs): `safe_limit >= lender_sanction`

"Why: Minimal fixed living expenses and low ongoing debt allow your safe borrowing capacity to match the maximum standard banking limit."

Output 3 "Why" (Fair Rate Band & APR)

- Trigger 1 (Asset Backed Pivot): `has_property_collateral == True`

"Why: Pledging an unencumbered shop/property shifts your product to Loan Against Property (LAP), reducing the rate from 18%+ unsecured to {min_rate}%-{max_rate}%."
- Trigger 2 (Prime Credit Standing): `cibil >= 750`

"Why: Your 750+ CIBIL qualifies you for the prime pricing band, while a {fee}% processing fee plus GST sets the true all-in APR at {apr}%."
- Trigger 3 (Unscored Bureau History): `cibil == "Unknown"`

"Why: A wider rate band of {min_rate}%-{max_rate}% is applied because your credit history is unverified, ensuring transparency without penalizing you as a defaulter."

Output 4 "Why" (Safe Monthly EMI Ceiling)

- Trigger 1 (Rent & Expense Constraint): `safe_emi < lender_foir_emi`

"Why: While bank FOIR norms permit up to ₹{lender_foir_emi:,}/month, your monthly rent of ₹{rent:,} and living expenses cap your safe payment at ₹{safe_emi:,}."
- Trigger 2 (Stress Test Downward Adjustment): `stress_test_adjusted == True`

"Why: Under a 20% income downturn, an EMI above ₹{safe_emi:,} would consume over 65% of your remaining cash flow, so the ceiling was adjusted downward."
- Trigger 3 (Regulatory 50% Cap): `lender_foir_emi <= safe_emi`

"Why: Capped at ₹{safe_emi:,} because regulatory guidelines prevent total monthly debt repayments from exceeding 50% of your take-home pay."

6. THE MOBILE NEGOTIATION CARD ENGINE

The Negotiation Card converts calculations into actionable counter-benchmarks that the borrower can reference during discussions with a loan officer.

MOBILE NEGOTIATION CARD	
1. TARGET ANCHORS	- Target Product: Prime Unsecured Corporate Loan - Fair Rate Band: 10.50% – 11.25%

• Hard APR Cap: 11.80% (Processing Fee ≤ 1.0% + GST) • Safe EMI Limit: ₹22,000 / month
2. BORROWER LEVERAGE POINTS • CIBIL score of 780 qualifies for prime corporate tier. • Tier-1 listed corporate employer salary account. • Existing FOIR < 20% leaves ample debt capacity.
3. BANK TRAPS TO REJECT • Mandatory single-premium credit shield insurance. • Processing fees exceeding 1.0% + GST. • Prepayment lock-ins exceeding 12 months.
4. VERBATIM SPOKEN COUNTER-SCRIPT "My CIBIL is 780 and my employer is Category-A. Standard corporate pricing clears between 10.5% and 11.25%. If the APR exceeds 11.80% with processing fees capped at 1%, I will move the sanction to my primary salary bank. Do not deduct loan insurance from disbursement."

6.1 Dynamic Assembly Logic

- **Target Product Identification:** Evaluates collateral assets and equipment utility flags to assign the appropriate borrowing vehicle (e.g., LAP for unencumbered real estate, Priority Asset Financing for EV commercial equipment, or Prime Corporate Personal Loan).
- **Leverage Extraction:** Surfaces verified strengths (e.g., CIBIL ≥ 750, stable operational vintage ≥ 3 years, unencumbered real estate, or operating cost savings from productive tools).
- **Trap Identification:** Identifies common retail lending markups based on the borrower's profile (e.g., bundled single-premium insurance, express high-rate unsecured business loans, or hidden processing charges).
- **Spoken Script Templating:** Compiles targets and leverage points into a conversational counter-statement:

"I am applying for a {Target Product}. Based on {Primary Leverage Point}, standard regulatory pricing indicates a fair rate of {Min Rate}% to {Max Rate}%, with an all-in APR cap of {All-In APR}%. My repayment ceiling is strictly ₹{Safe Monthly EMI}/month; I will not accept structured tenures, fee markups above {Fee Cap}, or mandatory single-premium insurance add-ons that breach these limits."

7. VERIFIED PERSONA TEST SUITE

The implementation must validate correctly against these three canonical test vectors:

Test Case 1: Priya (Salaried MNC Professional)

- **Inputs:** Net Income: ₹1,40,000/mo | Existing EMIs: ₹0 | Rent: ₹32,000/mo | Living Expenses: ₹25,000/mo | Target Loan: ₹8,00,000 | Tenure: 48 mos | CIBIL: 780 | Category: Salaried | Tier: Listed Tier-1 Corporate | Runway: 8 months.
- **Expected Calculations:**
 - Base FOIR: 55% (Max Bank EMI = ₹77,000)
 - Safe Buffer: 5% (₹7,000) → Borrower Safe EMI = 140000 – 32000 – 25000 – 7000 = ₹76,000
 - Requested Loan EMI (at 11.0% for 48 mos): ₹20,690/mo
 - All-In APR (with 1.0% fee + GST): 11.65%
- **Expected Outputs:**
 - *O*₁ (**Verdict**): BORROW
 - *O*₂ (**Limits**): Lender Sanction: ₹29,70,000 | Safe Limit: ₹29,30,000
 - *O*₃ (**Rate & APR**): Fair Rate: 10.50% – 11.25% | All-In APR: 11.65%
 - *O*₄ (**Monthly Ceiling**): ₹76,000/mo (Survives 20% income shock)
 - **Negotiation Card:** Target prime corporate rate; refuse credit-shield insurance bundling.

Test Case 2: Ravi (Self-Employed Kirana Store Owner)

- **Inputs:** Gross Cash Intake: ₹85,000/mo | Reported ITR Profit: ₹2,40,000/yr | Existing EMIs: ₹6,000 | Rent: ₹0 (Owned shop & residence) | Living Costs: ₹28,000/mo | Target Loan: ₹15,00,000 | Tenure: 84 mos | CIBIL: Unscored | Property Collateral: Unencumbered commercial shop valued at ₹45,00,000.
- **Expected Calculations:**
 - Traditional Unsecured ITR Cap: Caps unsecured loans at ~₹5,00,000 at 18%–22% APR.
 - Product Pivot (LAP at 50% LTV): Unlocks up to ₹22,50,000 at 9.75%–10.50%.

- Safe EMI Capacity: $85000 - 6000 - 28000 - 8500(\text{buffer}) = ₹42,500/\text{mo.}$
- Requested Loan EMI (₹15L at 10% for 84 mos): ₹24,900/mo.

• **Expected Outputs:**

- O_1 (**Verdict**): BORROW
- O_2 (**Limits**): Lender Sanction (LAP): ₹22,50,000 | Safe Limit: ₹21,00,000
- O_3 (**Rate & APR**): Fair Rate: 9.75% – 10.50% | All-In APR: 10.85%
- O_4 (**Monthly Ceiling**): ₹42,500/mo (Anchored to business low-season)
- **Negotiation Card**: Demand LAP pricing; reject 18%+ unsecured business installment loans.

Test Case 3: Anita (Informal Gig Economy Courier)

- **Inputs**: Monthly Earnings: ₹28,000/mo | Existing EMIs: ₹1,500 (Smartphone) | Living Costs: ₹18,000/mo | Rent: ₹5,000/mo | Target Loan: ₹60,000 | Tenure: 18 mos | Recent Bounces: 1 auto-debit bounce | Active App Loans: ₹15,000 across 3 apps at 36% APR | Asset Type: Commercial EV Scooter (Saves ₹3,500/mo petrol).

• **Expected Calculations:**

- Kill-Switch Active: `bounces_last_6m == 1` and `app_debt_apr > 24%`.
- Safe EMI Capacity (Pre-Restructuring): $28000 - 1500 - 5000 - 18000 = ₹3,500/\text{mo.}$
- Requested EV Loan EMI: ₹3,680/mo (Exceeds current surplus).

• **Expected Outputs:**

- O_1 (**Verdict**): DON'T BORROW
- O_2 (**Limits**): Lender Sanction: ₹0 (Blocked) | Safe Limit: ₹0
- O_3 (**Rate & APR**): Fair Rate: 12.0% – 13.5% (EV priority loan) | Current App Rate: 36%
- O_4 (**Monthly Ceiling**): ₹3,500/mo (Requires debt clearance before onboarding)
- **Negotiation Card**: Restructure existing app loans; leverage operational EV fuel savings to finance asset through OEM NBFC.

8. CODE STRUCTURE & IMPLEMENTATION ROADMAP

```

borrowing-copilot/
├── README.md           # Complete System Architecture & Documentation
├── app.py              # Presentation Layer & UI Implementation
├── requirements.txt     # Minimal dependencies (e.g., streamlit)
└── engine/
    ├── __init__.py
    ├── underwriting.py  # Deterministic Math, APR, FOIR, and Limits
    ├── confidence.py    # Confidence Progression & Question Matrix (104 rows)
    ├── explanations.py  # Deterministic Priority Trees & "Why" Generators
    └── negotiation.py   # Negotiation Card & Dynamic Script Generator
  
```

Implementation Steps for the Agent:

1. **Build engine/underwriting.py**: Implement the core financial functions: PMT amortization, Present Value calculations, iterative IRR solver for all-in APR, regulatory FOIR lookups, and stress-test simulation loops.
2. **Build engine/confidence.py**: Implement the 104-row mapped confidence progression lookup table, managing question sequencing and tracking the confidence score from 60% to 95%.
3. **Build engine/explanations.py**: Codify the priority-checking trees that evaluate mathematical outputs and return the corresponding one-sentence "Why" diagnostics.
4. **Build engine/negotiation.py**: Construct the dynamic Negotiation Card dictionary, populating target terms, borrower leverage points, bank traps, and interpolated counter-scripts.
5. **Build app.py**: Create the mobile-responsive user interface:
 - Interactive input form with dynamic conditional rendering.
 - Visual confidence gauge reflecting the 104-row progress matrix.
 - Paired two-folded result cards (O_1 to O_4).
 - Standalone, high-contrast mobile Negotiation Card.

6. **Validate Test Vectors:** Execute the unit test suite against Priya, Ravi, and Anita to confirm outputs match expected benchmarks.
